QUASI-COMPACT OPENS OVER A GALOIS EXTENSION ARE DEFINED OVER A FINITE SUBEXTENSION AND FIXED BY AN OPEN SUBGROUP

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and consists of the descent step and the group-theoretic observation; nothing is referred to the reader.

Editorial note. The counted spans carry 1 navigation stub and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-Galois-action-quasi-compact-open is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Descent of a quasi-compact open to a finitely generated subalgebra. Same chapter, Lemma lemma-descend-open, invoked at proof lines 891-893. Supplies the finite subextension and the descended open of part (1).
- The Galois group of the extension over a finite subextension is open in the full Galois group. Standard Galois theory, used at proof lines 893-895. Supplies part (2).

Source reference. Original lemma 0388: sources/stacks-varieties/pinned-original-source.tex, source lines 812–812. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. Let $\overline{k}$ be a (possibly infinite) Galois extension of k. Let $V \subset X_{\overline{k}}$ be a quasi-compact open. Then*

(1) *there exists a finite subextension $\overline{k}/k'/k$ and a quasi-compact open $V' \subset X_{k'}$ such that $V = (V')_{\overline{k}}$,*
(2) *there exists an open subgroup $H \subset Gal(\overline{k}/k)$ such that $\sigma(V) = V$ for all $\sigma \in H$.*

Proof. By Lemma 0388 there exists a finite subextension $k'/k \subset \overline{k}$ and an open $V' \subset X_{k'}$ which pulls back to V. This proves (1). Since $\mathrm{Gal}(\overline{k}/k')$ is open in $\mathrm{Gal}(\overline{k}/k)$ part (2) is clear as well. □

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